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# Applications of $\pi$-photon-induced transparency in two-frequency pulse electron paramagnetic resonance experiments

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An approach to pulse electron paramagnetic resonance (EPR) experiments which are based on two different resonance fields is introduced. Instead of using two microwave (mw) sources or a magnetic field jump, bichromatic pulses consisting of a transverse microwave field with frequency $\omega_{mw}$ and a longitudinal radio frequency field with frequency $\omega_{rf}$ are employed. Such bichromatic pulses excite a number of multiple photon transitions at frequencies $\omega_{mw}+k\omega_{rf}$ $(k\in Z)$. The $\pi$-photon-induced transparency phenomenon is used to select the required transitions. This approach is used in the stimulated soft electron spin echo envelope modulation and the four-pulse double electron-electron resonance experiments. The results obtained using the bichromatic pulse approach are in agreement with those obtained with the standard pulse EPR techniques. It is shown that applying bichromatic pulses is straightforward and advantageous in several respects. © *2004 American Institute of Physics.* [DOI: 10.1063/1.1633259]

## I. INTRODUCTION

In pulse electron paramagnetic resonance (EPR) spectroscopy a number of pulse sequences make use of two microwave (mw) frequencies or a magnetic field jump.[1] Among these are stimulated soft electron spin echo envelope modulation (SS-ESEEM)[2] and three- and four-pulse double electron–electron resonance (DEER)[3,4] experiments.

The first type of experiment, SS-ESEEM, is one of the ESEEM techniques which are used to determine weak hyperfine interactions in paramagnetic species in solids. SS-ESEEM is based on the use of soft (i.e., long and weak) mw pulses, which are selective and thus can not excite simultaneously allowed and forbidden EPR transitions of the same spin system. The modulation of the echo amplitude is induced by using pulses with two different mw frequencies $\omega_{mw}$ and $\omega_{mw}+\Delta\omega$, one resonant with an allowed, and another one resonant with a forbidden transition. It has been shown that SS-ESEEM provides the same information as three-pulse ESEEM which is based on nonselective mw pulses.[2] However, SS-ESEEM can be carried out with much lower mw power and has the advantage of being free of blindspots. Due to the narrow frequency range excited by the selective mw pulses, the experiment has to be repeated with different values of $\Delta\omega$ to obtain the whole spectrum. SS-ESEEM is the method of choice when mw power is limited and the use of short nonselective mw pulses is not feasible, as is usually the case in high-field EPR.

The second type of experiment, three- and four-pulse DEER, is used to determine distances between two paramagnetic centers in solids by measuring the electron–electron dipolar interaction.[3–7] In the original three-pulse DEER experiment,[3] a primary two-pulse echo sequence with pulses with frequency $\omega_{mw}$ is used to excite the spins in the paramagnetic centers of the first kind. Then, the effect of irradiating the spins in paramagnetic centers of the second kind with frequency $\omega_{mw}+\Delta\omega$ on the primary echo amplitude is observed. This approach suffers from the instrumental deadtime, which may be a considerable drawback in disordered systems. In four-pulse DEER,[4] the influence of the deadtime is eliminated, resulting in an increase in resolution and sensitivity. This is achieved by observing a refocused echo instead of the primary echo.

Recent studies on multiple photon transitions[8–10] indicate that a pulse with mw frequency $\omega_{mw}+\Delta\omega$ can be replaced by a so-called bichromatic pulse. The bichromatic pulses used in this work consist of a mw pulse with frequency $\omega_{mw}$ during which a radio frequency (rf) pulse of the same length and frequency $\omega_{rf}=\Delta\omega$ is applied *parallel* to the external magnetic field $(B_0)$. Right-hand circularly polarized photons with frequency $\omega_{mw}$ and angular momentum 1 relative to the $B_0$ field axis ($\sigma^+$-photons) are absorbed from the transverse mw field. In addition, linearly polarized photons with frequency $\omega_{rf}$ and zero angular momentum relative to the $B_0$ field axis ($\pi$-photons)[11,12] are simultaneously absorbed or emitted from the *longitudinal* rf field. Therefore, such a bichromatic pulse excites not only the single-photon transition at $\omega_{mw}$, but also a number of multiple photon transitions at frequencies $\omega_{mw}+k\omega_{rf}$ (sidebands), where $k$ is a signed integer, referring to the number of rf $\pi$-photons absorbed (positive $k$) or emitted (negative $k$). The transition amplitudes of these $(|k|+1)$-photon transitions are weighted with a Bessel function of the first kind.[9,10] A remarkable feature of bichromatic pulses is that the transition amplitude of a transition with frequency $\omega_{mw}+k\omega_{rf}$ can be selectively set to zero by adjusting the ratio between amplitude and frequency of the rf field. The physics behind

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this so-called $\pi$-photon-induced transparency[10] is a destructive interference of different multiple photon processes. It can be expressed analytically by Bessel functions, with transparencies found at their corresponding zero-crossings. This phenomenon opens the possibility of using bichromatic pulses to substitute the second mw frequency in two-frequency pulse EPR experiments. A bichromatic pulse which fulfills the transparency condition for $\omega_{\mathrm{mw}}$ does not drive the transition with this frequency, but it drives the multiple photon transitions with frequencies $\omega_{\mathrm{mw}}+k\omega_{\mathrm{rf}}$ with transition amplitudes that are different from zero.

In this work we demonstrate that bichromatic pulses can substitute the second mw source both in SS-ESEEM and four-pulse DEER experiments. The results are compared to those obtained using traditional techniques: bichromatic-pulse SS-ESEEM is compared to three-pulse ESEEM, and bichromatic-pulse DEER is compared to standard two-frequency DEER. The behavior of bichromatic pulses which fulfill the transparency or near-transparency condition is investigated, and recommendations for obtaining optimum results are given.

## II. THEORY

In this section we briefly review the theory developed in previous studies on multiple photon transitions.[9,10,13] For an $S=1/2$ electron spin system in a static magnetic field and irradiated by a bichromatic field consisting of a transverse mw field and a longitudinal rf field, the spin Hamiltonian is given by:

$$\mathcal{H}(t)=\omega_S S_z+2\omega_1\cos(\omega_{\mathrm{mw}}t)S_x+2\omega_2\cos(\omega_{\mathrm{rf}}t)S_z, \tag{1}$$

where $\omega_S=-\gamma_e B_0=g\beta_e B_0/\hbar$ is the electron Larmor frequency, $2\omega_1=-\gamma_e B_{\mathrm{mw}}$ and $\omega_{\mathrm{mw}}$ are the amplitude and frequency of the mw field, and $2\omega_2=-\gamma_e B_{\mathrm{rf}}$ and $\omega_{\mathrm{rf}}$ are the amplitude and frequency of the rf field. For brevity, in Eq. (1) we set the phases of the mw and rf field to zero.

The time-dependence of the oscillating mw field is removed by a transformation to the singly rotating frame (SRF) and neglecting the counter-rotating part of the mw field:

$$\mathcal{H}_{\mathrm{SRF}}(t)=[\omega_S-\omega_{\mathrm{mw}}]S_z+\omega_1 S_x+2\omega_2\cos(\omega_{\mathrm{rf}}t)S_z. \tag{2}$$

For a convenient description of the multiple photon processes, a transformation to a toggling frame,[9] a kind of a generalized rotating frame, is performed with the rotation operator

$$R(t)=\exp\left(i\left[k\omega_{\mathrm{rf}}t+\left(\frac{2\omega_2}{\omega_{\mathrm{rf}}}\right)\sin(\omega_{\mathrm{rf}}t)\right]S_z\right), \tag{3}$$

resulting in the Hamiltonian

$$\begin{aligned}\mathcal{H}_{TF,k}(t)=&(\omega_S-\omega_{\mathrm{mw}}-k\omega_{\mathrm{rf}})S_z\\&+\sum_{n=-\infty}^{\infty}J_n\left(\frac{2\omega_2}{\omega_{\mathrm{rf}}}\right)\omega_1 e^{i[(k+n)\omega_{\mathrm{rf}}t]S_z}\\&\times S_x e^{-i[(k+n)\omega_{\mathrm{rf}}t]S_z},\end{aligned} \tag{4}$$

where $k\in Z$, and $J_{\mathrm{n}}$ is a Bessel function of the first kind and order $n$.

In the sum in Eq. (4) only the term with $n=-k$ is time-independent and responsible for the multiple photon transition with the frequency $\omega_{\mathrm{mw}}+k\omega_{\mathrm{rf}}$. The time-dependent terms lead to a Bloch–Siegert-like shift

$$\Delta_k=\left(\frac{2\omega_1^2}{\omega_{\mathrm{rf}}}\right)\sum_{l\neq k}\frac{J_l^2(2\omega_1/\omega_{\mathrm{rf}})}{(k-l)}, \tag{5}$$

and the effective spin Hamiltonian in the toggling frame may be written as

$$\mathcal{H}_{TF,k}=([\omega_S+\Delta_k]-[\omega_{\mathrm{mw}}+k\omega_{\mathrm{rf}}])S_z+\omega_{1,k}S_x, \tag{6}$$

where $\omega_{1,k}$ is calculated in first order $(\omega_1/\omega_{\mathrm{rf}}\ll 1)$ as

$$\omega_{1,k}=\omega_1 J_{-k}\left(\frac{2\omega_2}{\omega_{\mathrm{rf}}}\right)\equiv\omega_1 J_{-k}(z). \tag{7}$$

The Hamiltonian in Eq. (6) essentially describes the spin system exposed to an effective radiation field with frequency $\omega_{\mathrm{mw}}+k\omega_{\mathrm{rf}}$ and amplitude $\omega_{1,k}$. Since the transition amplitude $\omega_{1,k}$ is zero for $J_k(z)=0$, at zero-crossings of the $k$-th Bessel function the spin system becomes transparent for the $(|k|+1)$-photon transition. At the same time, the transition amplitudes of all the other multiple photon transitions are different from zero. For a transition resonant with the mw frequency $\omega_{\mathrm{mw}}$ the first transparency occurs at $z\approx 2.4$ $(J_0(2.4)\approx 0)$. For this $z$-value the transition amplitude of the transitions with the two frequencies $\omega_{\mathrm{mw}}\pm\omega_{\mathrm{rf}}$ is $|\omega_{1,\pm 1}|=\omega_1|J_{\mp 1}(2.4)|\approx 0.52\omega_1$. This is the situation which is used in this work. Note, that at $z=2.4$ not only two-photon transitions with frequencies $\omega_{\mathrm{mw}}\pm\omega_{\mathrm{rf}}$ $(|J_{\pm 1}(2.4)|=0.52)$, but also three- and four-photon transitions with frequencies $\omega_{\mathrm{mw}}\pm 2\omega_{\mathrm{rf}}$ $(|J_{\pm 2}(2.4)|=0.43)$ and $\omega_{\mathrm{mw}}\pm 3\omega_{\mathrm{rf}}$ $(|J_{\pm 3}(2.4)|=0.19)$ can efficiently be driven. The efficiency of driving higher-order multiple photon transitions can be neglected.

## III. EXPERIMENT

All the experiments were carried out on commercial X-band EPR spectrometers (Bruker Elexsys E580 and E680). The rf field parallel to the static magnetic field was produced by the rf coil of a pulse ENDOR probehead (Bruker ER 4118X-MD5-EN) rotated by 90° around its axis. The rf power was generated by an arbitrary function generator (LeCroy LW420B) and then amplified by a broadband amplifier (Amplifier Research, models 250A250A and 500A100A). The phase of the rf field was locked with respect to the pulse sequence.[10] Quadrature detection was used in all the experiments.

The maximum rf field amplitude is limited to about 1.5 mT (value in the rotating frame corresponding to $\omega_2$). Therefore, for a frequency of 14 MHz (proton Larmor frequency at X-band) the maximum value for $z=2\omega_2/\omega_{\mathrm{rf}}$ is about $z=6$. For lower frequencies higher $z$ values can be achieved. This is sufficient for our experiments and we recommend to use the first transparency condition with $z=2.4$ for the transition resonant with $\omega_{\mathrm{mw}}$, and not higher zero-crossings of the Bessel function. This is because the efficiency to drive the transitions at $\omega_{\mathrm{mw}}\pm\omega_{\mathrm{rf}}$ is larger for the first transparency condition $(z=2.4)$ than for higher ones. For example, for

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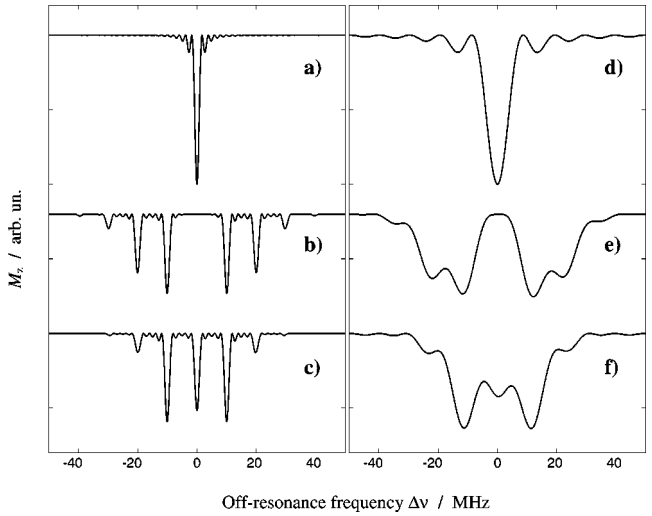



FIG. 1. Simulations of the $M_z$ magnetization of an $S=1/2$ spin system after a bichromatic pulse with a radio frequency of 10 MHz as a function of the off-resonance frequency $\Delta\nu$. (a)–(c) pulse length 500 ns and (d) and (f) 100 ns. The mw component of the bichromatic pulse has a nominal flip angle $\pi$. (a) and (d) $z=0$ (rf off). (b) and (e) $z=2.4$ (transparency condition fulfilled). (c) and (f) $z=1.5$ (transparency condition not fulfilled). Plots (a)–(f) use the same scale.

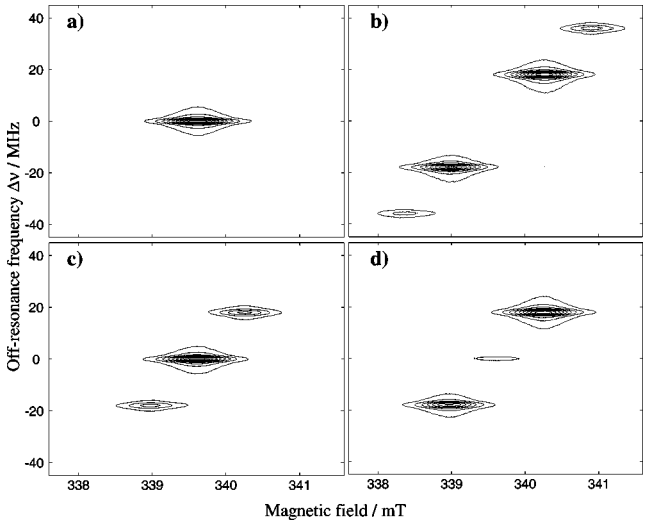



FIG. 2. Contour plots of the FID-detected EPR spectra of a coal after a 1-$\mu$s bichromatic pulse as a function of the static magnetic field (room temperature, $\omega_{mw}/2\pi=9.51$ GHz, $\omega_{rf}/2\pi=18$ MHz). (a) $z=0$ (no rf field is applied). (b) $z=2.4$ (transparency condition is fulfilled). (c) $z=1.5$ (transparency condition is not fulfilled). (d) $z=1.5$, the rf phase cycle [(0)-($\pi$)] is applied.

$z=2.4$ the transition amplitudes at $\omega_{mw}\pm\omega_{rf}$ are $|\omega_{1,\pm1}|=\omega_1|J_{\mp1}(2.4)|=0.52\omega_1$, while for $z=5.5$, corresponding to the second transparency condition (second zero-crossing of $J_0$), the transition amplitudes are reduced to $|\omega_{1,\pm1}|=\omega_1|J_{\mp1}(5.5)|=0.34\omega_1$.

An important requirement is that the bichromatic pulse should have a length of at least one period of the radio frequency, since otherwise the excitation profile of the rf pulse will be too broad. Thus, for a rf of 14 MHz the length of the bichromatic pulse should be 70 ns or longer.

In the bichromatic SS-ESEEM experiment we used a single crystal of $^{63}$Cu(II)-doped Zn(picolinate)$_2$ grown by slowly lowering the temperature of a saturated aqueous solution of Zn(picolinate)$_2$ and $^{63}$Cu(picolinate)$_2$ (Cu:Zn $=1:200$), and a powder of the same compounds with a ratio of about Cu:Zn$=1:100$. For the DEER experiments the nitroxide biradical (CAS 312624-83-21) in *o*-terphenyl (0.5 mg/g) was used. After melting ($\approx$60 °C) and shock-freezing by immersing the melt into liquid nitrogen, a glass was formed. To demonstrate the excitation profile of the bichromatic pulse, a coal sample was used.

## IV. RESULTS AND DISCUSSION

### A. Excitation profile of the bichromatic pulse

Before implementing the bichromatic-pulse approach for SS-ESEEM and DEER we demonstrate in simulations and experiment that a bichromatic pulse excites not only the transitions close to $\omega_{mw}$, but also transitions close to the frequencies $\omega_{mw}+k\omega_{rf}$, in particular $\omega_{mw}\pm\omega_{rf}$ (first sideband).

Figure 1 shows numerical simulations of the $M_z$ magnetization of the $S=1/2$ spin system after the bichromatic pulse with nominal flip angle $\pi$ of the mw field. Two pulse length [(a)–(c): 500 ns, (d)–(f): 100 ns] and three values of $z$ are considered. For $z=0$ [Figs. 1(a) and 1(d)], the mw pulse changes $M_z$ only for spins close to $\omega_{mw}$ (resonance offset $\Delta\nu=0$). If the transparency condition is fulfilled ($z=2.4$), no excitation occurs at $\omega_{mw}$, and only the spins with resonance frequencies $\omega_{mw}\pm\omega_{rf}$, $\omega_{mw}\pm2\omega_{rf}$, and $\omega_{mw}\pm3\omega_{rf}$ are excited [Figs. 1(b) and 1(e)]. The contribution of higher-order multiple photon transitions is negligible. Note, that for a 500-ns pulse the intensities of these excitation sidebands are in good agreement with the values of $J_{\pm k}(2.4)$ mentioned above. For the 100-ns pulse the excitation profiles of the sidebands overlap, but the excitation at $\omega_{mw}$ is still close to zero [Fig. 1(e)]. This means that a bichromatic pulse of proper length which fulfills the transparency condition can efficiently be used for broadband excitation with frequencies different from the mw frequency. If the transparency condition is not fulfilled, $z=1.5$ [Figs. 1(c) and 1(f)], transitions close to $\omega_{mw}$ and $\omega_{mw}+k\omega_{rf}$ are simultaneously excited, as it is expected from analytical consideration in Sec. II. Note, that for the transverse magnetizations $M_x$ and $M_y$ an analogous behavior is found, reflecting the excitation at frequencies $\omega_{mw}+k\omega_{rf}$ by the bichromatic pulse.

Figure 2 shows free induction decay (FID)-detected EPR spectra of a coal, recorded at X-band with a 1-$\mu$s bichromatic pulse, as a function of static magnetic field $B_0$. For $z=0$, i.e., when only a mw pulse is applied [Fig. 2(a)], only the spins with resonance frequencies close to $\omega_{mw}$ are excited (frequency offset $\Delta\nu=0$ and $B_0=339.6$ mT). If the transparency condition is fulfilled for the bichromatic pulse [$z=2.4$, Fig. 2(b)], the spins at $\Delta\nu=0$ are no longer excited, but peaks at $\Delta\nu=\pm18$ MHz and $\Delta\nu=\pm36$ MHz are observed, corresponding to two- and three-photon transitions at $\omega_{mw}\pm\omega_{rf}$ and $\omega_{mw}\pm2\omega_{rf}$, respectively. The field positions of these peaks are shifted by $\pm0.64$ mT (18 MHz) and $\pm1.28$ mT (36 MHz) in agreement with the shift of the resonance frequency. Higher-order sidebands are not observed, since they have much lower intensities [see Fig. 1(b)].

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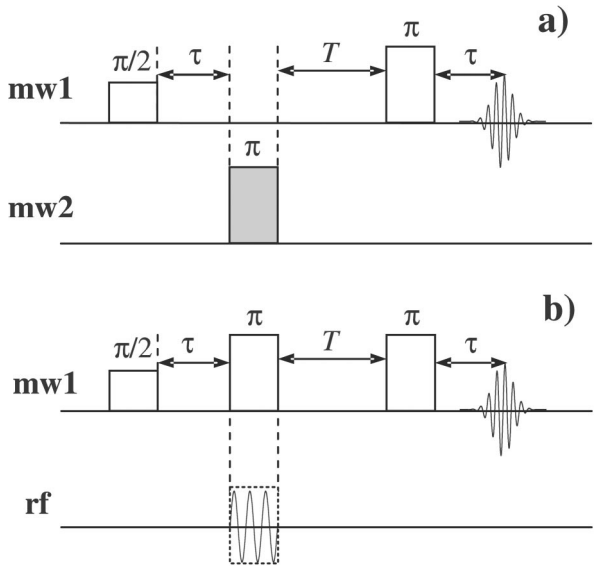



FIG. 3. Pulse schemes for (a) two-frequency SS-ESEEM and (b) bichromatic-pulse SS-ESEEM.

If the transparency condition for the bichromatic pulse is not fulfilled [$z=1.5$, Fig. 2(c)], the spins close to $\omega_{\mathrm{mw}}$ ($\Delta\nu=0$) and $\omega_{\mathrm{mw}}\pm\omega_{\mathrm{rf}}$ ($\Delta\nu=\pm\omega_{\mathrm{rf}}/2\pi$) are simultaneously excited. The peak at $\Delta\nu=0$ representing single-photon transitions can be eliminated by using the rf phase cycle [(0)-($\pi$)], as shown in Fig. 2(d).[10] In this case imperfections of the phase cycling lead to a small residual peak at $\omega_{\mathrm{mw}}$ ($\Delta\nu=0$). It is therefore superior to remove the excitation at $\omega_{\mathrm{mw}}$ directly by using $\pi$-photon-induced transparency. However, both methods can be combined for obtaining the best results.

Note, that both the simulations in Fig. 1 and the experimental results in Fig. 2 demonstrate the same properties of bichromatic pulses: the excitation of the sidebands with frequencies $\omega_{\mathrm{mw}}+k\omega_{\mathrm{rf}}$, and the possibility to eliminate the excitation at the central band using $\pi$-photon-induced transparency.

## B. Stimulated Soft ESEEM

### 1. Pulse sequence and bichromatic pulses

The pulse sequence used in the SS-ESEEM experiment[2] is shown in Fig. 3(a). The first pulse with mw frequency $\omega_{\mathrm{mw1}}$ creates electron coherence on an allowed (forbidden) transition. The second pulse with mw frequency $\omega_{\mathrm{mw2}}$ is resonant with a forbidden (allowed) transition and transfers the electron coherence to nuclear coherence, which evolves during time $T$. The third pulse again with mw frequency $\omega_{\mathrm{mw1}}$ transfers the nuclear coherence back to forbidden (allowed) electron coherence, which refocuses to a coherence-transfer echo.[1] Time $T$ is incremented, and the modulation of the echo amplitude is monitored. For this purpose the coherence-transfer echo can be demodulated and then integrated, or the absolute value of the quadrature-detected echo can be computed.

SS-ESEEM with a bichromatic pulse is shown in Fig. 3(b). The second pulse with frequency $\omega_{\mathrm{mw2}}$ is replaced by a bichromatic pulse. The general features of SS-ESEEM are discussed in detail in Ref. 2, here we concentrate on the characteristics of the experiments with a bichromatic pulse. To find the transparency condition for the bichromatic pulse, the pulse sequence $(\pi/2)_{\mathrm{bichrom}}-\tau-\pi_{\mathrm{mw}}-\tau-$echo, was used.[10] The rf amplitude $2\omega_2$ is incremented from zero up to the value where the echo vanishes. However, in case of SS-ESEEM it is sufficient just to run the bichromatic-pulse SS-ESEEM sequence and follow the echo after the second pulse, which should vanish at $z=2.4$. The effect of the first two pulses then corresponds to the sequence $(\pi/2)_{\mathrm{mw}}-\tau-\pi_{\mathrm{bichrom}}-\tau-$echo, which can also be used to find the transparency condition.

Note, that for each sideband the bichromatic pulse excites the transitions with two frequencies, $\omega_{\mathrm{mw}}+|k|\omega_{\mathrm{rf}}$ and $\omega_{\mathrm{mw}}-|k|\omega_{\mathrm{rf}}$. In the case of SS-ESEEM and DEER this is an advantage, since a larger number of spins is simultaneously excited which improves sensitivity.

### 2. Single crystal measurements

The potential of bichromatic-pulse SS-ESEEM is demonstrated on a single crystal of $^{63}$Cu(II)-doped Zn(picolinate)$_2$.[14,15] All the spectra are measured at a temperature of 20 K with an arbitrary crystal orientation. The lines in the echo-detected EPR spectrum have a width of 0.5–0.7 mT (fwhh), corresponding to 14–20 MHz, which is close to the proton Zeeman frequency at X-band. Allowed EPR transitions are thus preferably situated close to the line maximum, while the forbidden transitions are found in the wings of the lines. We can therefore choose the frequencies $\omega_{\mathrm{mw1}}$ and $\omega_{\mathrm{rf}}$ for the following two cases: (1) $\omega_{\mathrm{mw1}}$ close to the allowed transitions, and $\omega_{\mathrm{mw1}}\pm\omega_{\mathrm{rf}}$ close to the forbidden transitions, or (2) $\omega_{\mathrm{mw1}}$ close to the forbidden transitions, and $\omega_{\mathrm{mw1}}+\omega_{\mathrm{rf}}$ or $\omega_{\mathrm{mw1}}-\omega_{\mathrm{rf}}$ close to the allowed transitions. We verified, that the experiment can be carried out for both cases (1) and (2). However, if the available mw power is the limiting factor, the second approach is superior, since the transition amplitude of a transition at $\omega_{\mathrm{mw1}}\pm\omega_{\mathrm{rf}}$ is about a factor of two lower at $z\sim2.4$, compared to a monochromatic pulse [see Eq. (7), $|J_{\pm1}(2.4)|=0.52$]. It is therefore reasonable to set one of the frequencies $\omega_{\mathrm{mw1}}\pm\omega_{\mathrm{rf}}$ on-resonant with the allowed transitions, and the frequency $\omega_{\mathrm{mw1}}$ on-resonant with the forbidden transitions. The optimum of the echo intensity can then be achieved by increasing the amplitudes of the first and the third pulse.[2]

Figure 4 shows electron coherence-transfer echoes observed in the SS-ESEEM experiments with a bichromatic pulse and $z$ close to the transparency condition. At $z=2.4$ the transition amplitude for transitions at frequency $\omega_{\mathrm{mw1}}$ is zero, and only the transitions at $\omega_{\mathrm{mw1}}\pm\omega_{\mathrm{rf}}$ are driven. As a result only the electron coherence-transfer echo is observed. If the transparency condition is not exactly fulfilled, the observed signal is a superposition of a residual echo from the three-pulse sequence and the coherence-transfer echo. The phase of the residual three-pulse echo changes sign when $z$ crosses the value 2.4, in agreement with the behavior of $J_0$, while the phase of the coherence-transfer echo remains the same, as $J_{\pm1}$ does not change sign in this range of $z$ values.

Figure 5(a) shows the three-pulse ESEEM spectrum (between 8 and 20 MHz) obtained with short nonselective mw pulses of length 20 ns. It consists of several proton peaks

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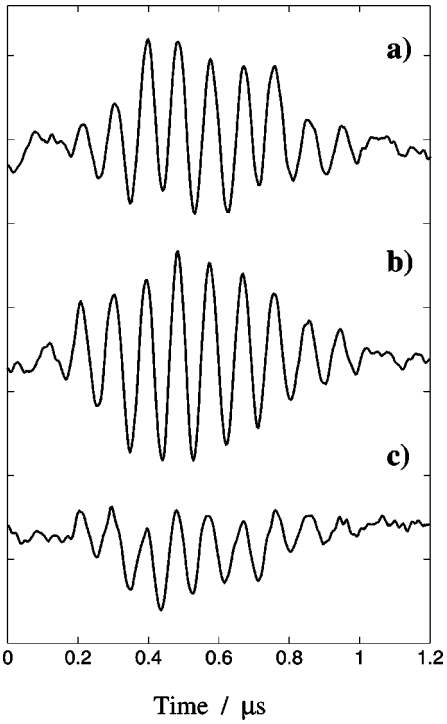



FIG. 4. Coherence-transfer echoes measured with the bichromatic-pulse SS-ESEEM sequence on a single crystal of $^{63}$Cu(II)-doped Zn(picolinate)$_2$ recorded as a function of the time after the last pulse for three different values of $z=2\omega_2/\omega_{rf}$. (a) $z<2.4$, (b) $z=2.4$ (transparency condition fulfilled), and (c) $z>2.4$.

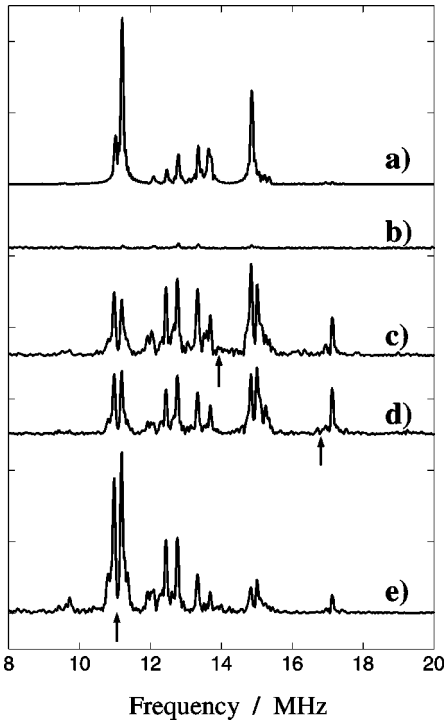



FIG. 5. Comparison of three-pulse ESEEM and bichromatic-pulse SS-ESEEM experiments on a single crystal of $^{63}$Cu(II)-doped Zn(picolinate)$_2$, arbitrary orientation, $T=20$ K. (a) Three-pulse ESEEM spectrum obtained with mw pulses of length 20 ns, (b) corresponding spectrum obtained with mw pulses of length 400 ns, (c)–(e) bichromatic-pulse SS-ESEEM spectra obtained with pulses of length 400 ns and radio frequencies (c) 14 MHz, (d) 17 MHz, and (e) 11 MHz marked by arrows. Plots (b)–(e) use the same scale.

close to the proton Larmor frequency of 12.7 MHz. Figures 5(b)–5(e) demonstrate the superiority of SS-ESEEM over three-pulse ESEEM, when three long low-power mw pulses (400 ns) are used. Figure 5(b) shows the standard three-pulse ESEEM spectrum, and Figs. 5(c)–5(e) the SS-ESEEM spectra measured by using bichromatic pulses with different radio frequencies (marked by arrows). The improvement in signal/noise ratio is tremendous. Note that we used pulses with a length of 400 ns to achieve the conditions where three-pulse ESEEM is no longer operating, but SS-ESEEM still is. This is because the bandwidth of a pulse of length 400 ns is by far too small to simultaneously excite allowed and forbidden transitions. In practice, pulses of about 150 ns are used in SS-ESEEM experiments.

The line positions in three-pulse ESEEM [Fig. 5(a)] and SS-ESEEM with a bichromatic pulse [Figs. 5(c)–5(e)] are the same. Artifacts at the radio frequency $\omega_{rf}$ are not observed. Some of the lines are more intense in the bichromatic-pulse experiment than in three-pulse ESEEM, because SS-ESEEM does not suffer from blindspots. In all bichromatic-pulse SS-ESEEM experiments shown in Fig. 5 the transparency condition was fulfilled for each value of the radio frequency. As expected, the line intensities are different in the three SS-ESEEM spectra. This is because the selective bichromatic pulse excites a narrow band of frequencies close to $k\omega_{rf}$. The closer a transition frequency is to $k\omega_{rf}$, the larger is the line intensity. This is nicely demonstrated in Figs. 5(c) and 5(d), and especially in Fig. 5(e), and is a well-known feature of SS-ESEEM.[2]

The fulfillment of the transparency condition is not critical for bichromatic-pulse SS-ESEEM. If the transition at $\omega_{mw}$ is not fully transparent, the residual echo created by the three-pulse sequence mentioned above is not disturbing, because it can be filtered out by a proper baseline correction. It is more relevant to choose the $z$ value such that the amplitudes of the transitions with frequencies $\omega_{mw1}\pm\omega_{rf}$ are maximum. This is the case for $z\sim2.0$, which is close to the transparency condition $z\sim2.4$. The experimentally observed spectra for the values in the range $z=1-3$ have all approximately the same signal intensities. The signal intensity in two-frequency SS-ESEEM is proportional to $\sin(\beta^{(2)}/2)$, where $\beta^{(2)}$ is the nominal flip angle of the second pulse.[2] Thus, taking into account only the contribution from the first sideband ($\omega_{mw1}\pm\omega_{rf}$), the signal in bichromatic-pulse SS-ESEEM should be proportional to $\sin[J_1(z)\beta^{(2)}/2]$. The experimental signal intensity versus $z$ is shown in Fig. 6 (quadrature detection and magnitude spectra calculation were used). It is in good agreement with the function $|A\sin(J_1(z)\beta^{(2)}/2)|$. As expected, minimum signal intensity is observed for $z\sim0$ and $z\sim4$. The finding that the signal intensity does not approach zero for $z\sim3.8$ is probably due to the contribution of the sidebands $\omega_{mw1}\pm2\omega_{rf}$ ($J_{\pm2}$) and $\omega_{mw1}\pm3\omega_{rf}$ ($J_{\pm3}$), which are not zero for $z\sim3.8$, but are close to the maxima of their corresponding amplitudes.

Thus, all the observed features agree well with the theoretical predictions. Bichromatic-pulse SS-ESEEM is easy to

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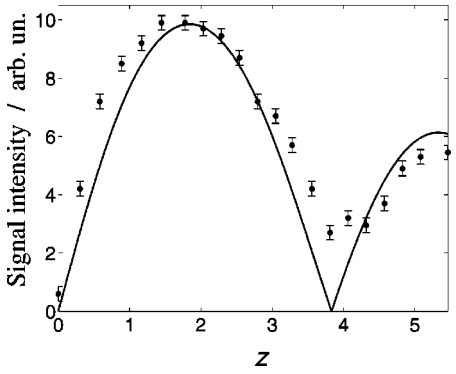



FIG. 6. Signal intensity in bichromatic-pulse SS-ESEEM spectra of a single crystal of $^{63}$Cu(II)-doped Zn(picolinate)$_2$ as a function of $z$. Experiment (circles) and function $|A \sin(J_1(z)\beta^{(2)}/2)|$ (line). A value $\beta^{(2)}=2.2$ for the nominal flip angle and a normalization factor $A=16.5$ are used for the simulated curve. Error bars correspond to the noise amplitude in the experimental spectra.

implement and does not require a precise tuning of the bichromatic pulse to the transparency condition. In fact, the best recipe for such measurements is to find the first zero-crossing of $J_0$, corresponding to $z=2.4$, and then set $z$ to 2.

### 3. Powder measurements

We now describe the measurements on a powder sample of $^{63}$Cu(II)-doped Zn(picolinate)$_2$. The main difference compared to the single crystal measurements is that the magnetic field position can no longer be set to select allowed or forbidden EPR transitions. Both types of transitions are excited simultaneously by all the pulses. The measurements were done at a temperature of 20 K and a magnetic field corresponding to an arbitrary (not single-crystal-like) orientation.

Figure 7(a) and 7(b) show three-pulse ESEEM spectra recorded with nonselective (20 ns) and selective mw pulses (200 ns), respectively. Bichromatic-pulse SS-ESEEM spectra recorded with 200-ns pulses are shown in Figs. 7(c)–7(e). Figures 7(b)–7(e) (shown at the same scale) manifest the large increase in signal intensity found in bichromatic-pulse SS-ESEEM.

The selectivity of the excitation in the SS-ESEEM experiment is demonstrated for different radio frequencies [Figs. 7(c)–7(e)]. Transitions close to the radio frequency are more intense than transitions far apart from $\omega_{\mathrm{rf}}$. However, despite the selectivity of the pulses the intensity variations are not very pronounced and the whole spectrum (up to 30 MHz) can be obtained with a single radio frequency. This is because the bichromatic pulse excites not only transitions close to $\omega_{\mathrm{mw}} \pm \omega_{\mathrm{rf}}$, but also transitions close to the frequencies $\omega_{\mathrm{mw}} + k\omega_{\mathrm{rf}}$, which is a further advantage of bichromatic-pulse SS-ESEEM compared to the version with two mw frequencies. Indeed, the bichromatic pulse has a much wider excitation bandwidth ($\pm\omega_{\mathrm{rf}}$, $\pm 2\omega_{\mathrm{rf}}$, $\pm 3\omega_{\mathrm{rf}}$, etc.), as is shown in Fig. 1, so that with a proper choice of pulse length and $\omega_{\mathrm{rf}}$, a wide range of the spectrum can be excited with a single radio frequency.

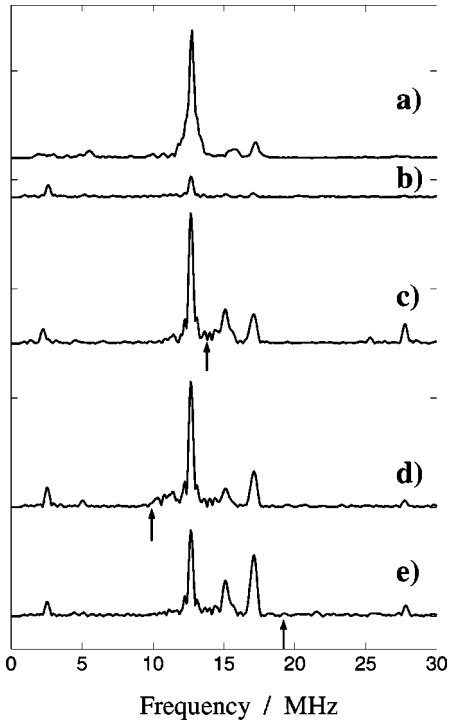



FIG. 7. Comparison of three-pulse ESEEM and bichromatic-pulse SS-ESEEM on a powder of $^{63}$Cu(II)-doped Zn(picolinate)$_2$. (a) Three-pulse ESEEM spectrum obtained with mw pulses of length 20 ns, (b) corresponding spectrum obtained with mw pulses of length 200 ns, (c)–(e) bichromatic-pulse SS-ESEEM spectra obtained with pulses of length 200 ns and radio frequencies (c) 14 MHz, (d) 10 MHz, and (e) 19 MHz marked by arrows. Plots (b)–(e) use the same scale.

## C. Four-pulse DEER

### 1. Pulse sequence and bichromatic pulses

The sequence for the four-pulse DEER experiment with two mw frequencies[1,3] is shown in Fig. 8(a). It consists of three pulses with frequency $\omega_{\mathrm{mw1}}$ and one pulse with frequency $\omega_{\mathrm{mw2}}$. The first pulse with flip angle $\pi/2$ and fre-

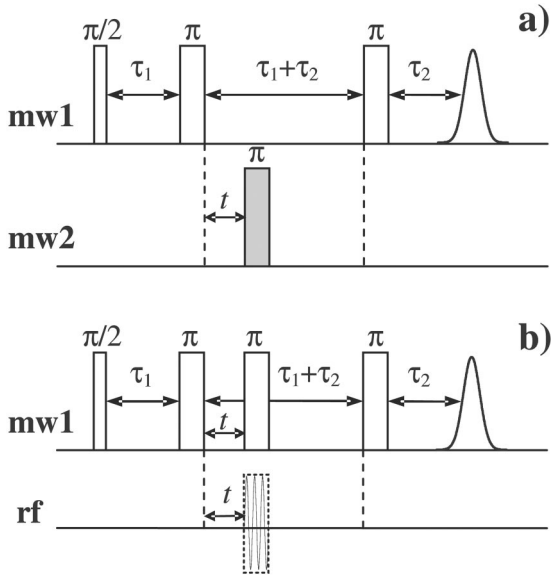



FIG. 8. Pulse schemes for (a) two-frequency four-pulse DEER and (b) four-pulse DEER with a bichromatic pulse.

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quency $\omega_{mw1}$ creates electron coherence. The second pulse with flip angle $\pi$ and frequency $\omega_{mw1}$ refocuses the coherence to an electron spin echo, which is formed at time $\tau_1$ after the pulse, and the third pulse with frequency $\omega_{mw1}$ and flip angle $\pi$ refocuses the coherence to a refocused echo. After the second pulse a mw pulse with flip angle $\pi$ and frequency $\omega_{mw2}$ is applied. The effect of this pulse on the refocused echo intensity is recorded by varying time $t$ between 0 and $2\tau_1$ ($\tau_1 \leqslant \tau_2$).

Figure 8(b) shows the corresponding sequence for the four-pulse DEER experiment with a bichromatic pulse. Instead of using a pulse with a second mw frequency, a bichromatic pulse with mw frequency $\omega_{mw1}$ and radio frequency $\omega_{rf}$ is applied.

A requirement for the DEER experiment is that the excitation profiles of the pulses with frequencies $\omega_{mw1}$ and $\omega_{mw2}$ do not overlap, otherwise the spectrum is distorted and intensity is reduced. Consequently, the bichromatic pulse has to be transparent for the transition at frequency $\omega_{mw1}$. The radio frequency should be set as high as possible to avoid an overlap of the excitation profiles. Another possibility is to use softer pulses, which allow one to reduce the radio frequencies. For typical radio frequencies of 10–25 MHz, one has any way to use a soft bichromatic pulse with a length of at least one period of the radio frequency. This leads to some decrease in signal intensity, since the number of spins involved in the experiment is reduced.

In the DEER experiment with a bichromatic pulse phase cycling has to be used when the transparency condition is not properly fulfilled. Three unwanted two-pulse echoes caused by the mw part of the bichromatic pulse and the three mw pulses have to be removed. To this end the phase cycle $[(0)\text{-}(\pi)]$ is applied to the first and second mw pulse and the mw part of the bichromatic pulse. Moreover, the rf phase cycle $[(0)\text{-}(\pi)]$ removes all imperfections of the rf pulse.

### 2. Experimental results

All DEER experiments have been carried out with the nitroxide biradical shown in Fig. 9 at a temperature of 50 K. This allows for a satisfactory tradeoff between a sufficiently long phase memory time and a longitudinal relaxation time, which is short enough to allow for a high repetition rate. The magnetic field was set to a value corresponding to the maximum of the echo intensity.

This nitroxide biradical was studied previously using the standard four-pulse DEER scheme with short nonselective pulses ($t_{\pi/2}=16$ ns, $t_\pi=32$ ns).[4] Figure 9(a) shows the time-trace obtained with the standard four-pulse DEER scheme with mw pulses of length $t_{\pi/2}=48$ ns and $t_\pi=100$ ns and a frequency difference $|\omega_{mw1}-\omega_{mw2}|/2\pi \equiv \Delta\omega_{mw}/2\pi = 24$ MHz, which is the same as in the experiment reported in Ref. 4. The time-trace obtained using four-pulse DEER with a bichromatic pulse ($t_{\pi/2}=48$ ns, $t_\pi=100$ ns, $\omega_{rf}/2\pi=24$ MHz) and a $z$ value close to the transparency condition is shown in Fig. 9(b). The two-frequency and bichromatic-pulse time-traces and their intensities are very similar. The bichromatic pulse again excites multiple photon transitions at several frequencies $\omega_{mw}+k\omega_{rf}$, which improves the signal intensity. On the other hand, some loss in sensitivity is expected for the bichromatic-pulse experiment compared to standard DEER because softer pulses are used, and for the bichromatic pulse the transparency condition cannot be exactly fulfilled for all excited spins.

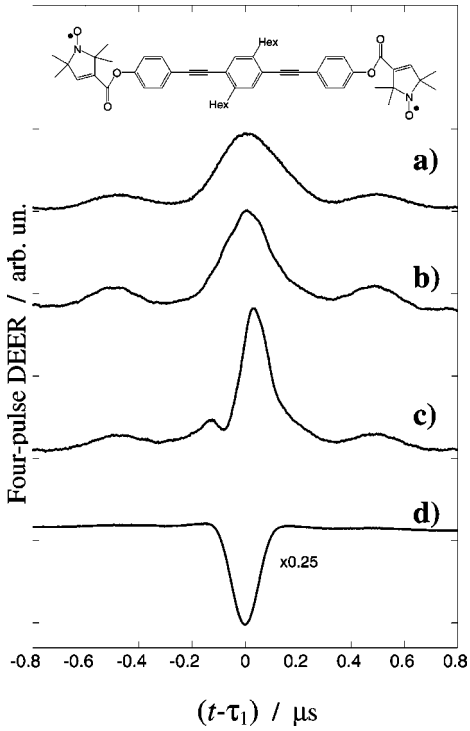



FIG. 9. Comparison of four-pulse DEER time-traces obtained with two mw frequencies and with the bichromatic-pulse schemes on the nitroxide biradical CAS 312624-83-21 shown on top. (a) Standard four-pulse DEER with two mw frequencies, $\Delta\omega/2\pi=24$ MHz, (b)–(d) four-pulse DEER with a bichromatic pulse with $\omega_{rf}/2\pi=24$ MHz and different $z$ values. (b) $z\approx 2.4$ (close to transparency condition), (c) $z\approx 1.5$, and (d) $z\approx 0.7$. All the plots use the same scale, the number of accumulations was the same in all experiments, the time increment was 4 ns.

The bichromatic-pulse DEER time-traces are sensitive to the value of $z$. If $z$ significantly deviates from the optimum value $z=2.4$, the time-trace gets distorted and a loss of the relevant signal is observed [Figs. 9(c) and 9(d)]. The time-trace in Fig. 9(c) ($z\sim 1.5$) still shows oscillations, but the central part is distorted. The oscillations in the time-trace in Fig. 9(d) ($z\sim 0.7$) are very weak and the central part is even inverted. This is because for a $z$ value far from transparency, the mw part of the bichromatic pulse inverts the two-pulse echo that appears after the second mw pulse.

Figure 10 shows the Fourier transforms of the time-traces in Figs. 9(a) and 9(b). Since our main goal is to compare these two experiments, a simple processing procedure was employed. The modulation of the echo amplitude corresponds to the intramolecular dipole–dipole coupling and reflects the electron–electron distance, while the exponential decay corresponds to the intermolecular coupling of electrons located at different molecules. The intermolecular part was eliminated by subtracting the fitted exponential decay. Then, the cosine spectrum was obtained by Fourier transformation. The two spectra are very similar and are close to the one presented in Ref. 4. The slight distortions in the bichromatic-pulse spectrum have no influence on the peak

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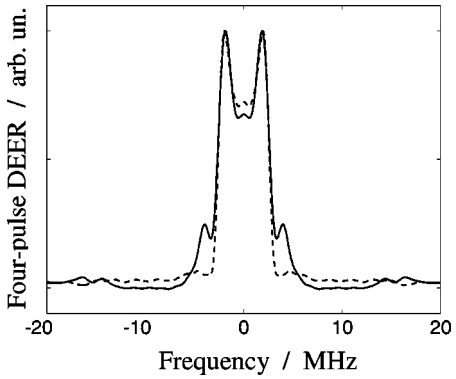



FIG. 10. Four-pulse DEER spectra of a nitroxide biradical. Dashed line: two-frequency DEER with $\Delta\omega/2\pi = 24$ MHz [Fig. 9(a)]; solid line: DEER with a bichromatic pulse, $\omega_{rf}/2\pi = 24$ MHz [Fig. 9(b)], and $z \sim 2.4$.

positions from which the value of the dipolar interaction $\omega_{DD}/2\pi = 2.0 \pm 0.5$ MHz is determined. This value is in good agreement with $\omega_{DD}/2\pi = 2.29 \pm 0.49$ MHz reported in Ref. 4. We suppose that the small peaks at about 4 MHz that show up in bichromatic-pulse spectrum are caused by the wide excitation bandwidth of the bichromatic pulse and represent the parallel components of the dipole interaction tensor.

Thus, the new four-pulse DEER experiment with a bichromatic pulse has proven to be an alternative for the standard four-pulse DEER approach with two mw frequencies. Both the three-pulse and four-pulse DEER experiments can be carried out with bichromatic pulses, a scheme which does not require a second mw source, and thus can often be experimentally easier to implement than the standard approach.

## V. CONCLUSIONS

In this work it is demonstrated that bichromatic pulses consisting of a transverse mw field and a longitudinal rf field can substitute the second mw frequency in two-frequency pulse EPR experiments. These results are in agreement with previous theoretical and experimental investigations.[8–10]

The use of a bichromatic pulse in stimulated soft ESEEM is straightforward and has some advantages compared to the standard experiment. The fulfillment of the transparency condition is not critical. Theory, simulations, and experiments show that the bichromatic pulse excites a much wider frequency range than a soft mw pulse of the same length. This compensates for the selectivity caused by the longer pulses used in bichromatic-pulse SS-ESEEM and is of advantage, since a wider range of the spectrum can be excited in a single experiment. It has been demonstrated that the bichromatic version of SS-ESEEM works well both for single crystals and powders.

The application of a bichromatic pulse in the four-pulse DEER sequence results in spectra very similar to those obtained with the standard four-pulse DEER scheme with two mw frequencies. In the case of DEER, it is important that the bichromatic pulse is transparent for the transition at $\omega_{mw1}$. In the approach here, rather long pulses have to be used. This may lead to some loss in signal intensity compared to the standard four-pulse DEER with short mw pulses. On the other hand, transitions at frequencies $\omega_{mw} + k\omega_{rf}$ are simultaneously excited, so that the signal intensity may be even higher than the one of the standard four-pulse DEER experiment with soft mw pulses.

Thus, this study shows that the bichromatic-pulse approach can be the useful tool for future applications in two-frequency pulse EPR experiments.

## ACKNOWLEDGMENTS

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